

Aspects of the Interaction between Cartography and Antecedentless Silent Elements*

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1. Introduction

Work on cartographic hierarchies¹ can in some cases be affected in significant ways by work on silent elements, in particular by the antecedentless kind to be discussed in this paper. In what follows, I will try to show in a number of cases what exactly I mean by this.

2. Antecedentless silent elements

Let me begin with an example taken from the game of baseball, which is standardly divided into nine parts, called innings, each of which is in turn divided into two parts, called the top and bottom of that inning. A banal sentence would be:

(1) Our team won the game with two home runs in the seventh inning.

Of initial syntactic interest is the fact that *inning* can remain silent:

(2) Our team won the game with two home runs in the seventh.

In a way familiar from English compound-like phrases, the following is possible as an alternative to (1):

(3) Our team won the game with two seventh inning home runs.

But in this case *inning* cannot remain silent:

(4) *Our team won the game with two seventh home runs.

Clearly, this is a syntactic fact, one that recalls, against the background of *Let's work hard today*, the following contrast that holds with regard to VP-deletion/silent VPs:

(5) Yes, it would be a good idea to.

(6) *Yes, to would be a good idea.

In both (4) and (6), the silent element is on a left branch in a way that it is not in (2) and (5). Setting aside for the moment the specific left branch question, let us conclude simply that an antecedentless silent element such as INNING in (2) (capitals will indicate silence) shares with the silent WORK HARD TODAY of (5) two key properties: Both are present in the syntax, and both are subject to configurational restrictions of a syntactic sort.

That the notion 'left branch' is at issue is, on the other hand, not clear. This can be seen by comparing (6) with:

(7) Your desire to has surprised just about everybody.

Correspondingly, the deviance of (4) contrasts with the acceptability of:

(8) Our team won the game with two top of the seventh home runs.

in a way that again emphasizes that we are looking at a syntactic restriction on what must be a syntactically present silent element (that in the case of INNING in these examples is antecedentless).

¹Cf. Cinque (1999), Rizzi (1997) and much subsequent work.

3. A possible account of the restriction

As for the question of what underlies the contrast between (4) and (8), in both of which INNING is contained within a left branch, it may be that the key is the presence in (8) of *the* preceding *seventh*. This is so as follows. The ordinal *seventh* in (1) is, unsurprisingly, preceded by *the*. Somewhat surprisingly, though, *seventh* is not preceded by *the* in (3), and in fact cannot be:

(9) Our team won the game with two (*the) seventh inning home runs.

Since (1) and (3) seem to have identical interpretations, and since ordinals cannot normally have no article in the way seen in:

(10) *The game is now in seventh inning.

it is natural to propose that (3)/(9) contains a silent counterpart of *the*, as indicated in:

(11) Our team won the game with two THE seventh inning home runs.

If so, then the unacceptable (4) must be as in:

(12) *Our team won the game with two THE seventh INNING home runs.

with two (antecedentless) silent elements that are syntactically close to one another. But having two silent elements that close to one another may in general be disallowed, thinking of the following well-known restriction concerning English VP-deletion and reduced auxiliaries:²

(13) Mary and John have travelled there and we have, too.

(14) *Mary and John have travelled there and we've, too.

In (14), we have a silent VP, but arguably also a close-by silent piece of *have*, if *have* = *ha+ve*, with the *ha-* part silent.³

From this perspective, the contrast between (14) and the following:

(15) I don't wanna.

(16) You really shoulda.

is arguably attributable to the fact that the reduced prepositions *to* in (15) and *of* in (16)⁴ are monomorphemic. (It may, finally, be that the interference effect seen in (12) and (14) is to be understood in terms of the movement approach to silent elements given in Kayne (2006).)

4. Interpretation; YEAR.

The above proposal concerning (2), illustrated in:

(17) ...in the seventh INNING.

straightforwardly assimilates (2) to (1), which contains:

(18) ...in the seventh inning.

and in so doing simplifies the semantics, in the sense that the interpretation of (2) simply reduces to that of (1).

The same holds for the proposal made in Kayne (2003), which attributed to a sentence like:

(19) At the age of seven, little Mary could already speak three languages.

²Cf. King (1970). We could equally well call 've et al. clitic auxiliaries, as in Thoms et al. (2019).

³As in the course that Michal Starke and I jointly taught at NYU in Fall 2002 on the morphology/syntax border; cf. also Collins (2020) and Collins and Kayne (2023).

⁴Cf. Kayne (1997).

the presence of silent YEAR(S):

(20) At the age of seven YEAR(S),...

and in so doing straightforwardly reduced the interpretation of (19) to that of:

(21) (?)At the age of seven years, little Mary could already speak three languages.

This silent YEAR(S)⁵ is, now unsurprisingly, subject to syntactic configurational restrictions that, at least to some extent, resemble those seen for INNING. For example, one has:

(22) Mary has a one-year old brother.

but not, with YEAR:

(23) *Mary has a one-old brother.

as opposed to:

(24) Mary has a little brother who's just turned one.

in which YEAR is legitimate.

Also clearly syntactic in nature is the striking comparative fact that (19) has no exact counterpart in either French or Italian:

(25) A l'âge de sept *(ans), Marie... (French 'at the-age of seven (years), Mary...')

(26) All'età di sette *(anni), Maria... (Italian - same)

Both French and Italian require their counterpart of *years* to be pronounced in such sentences. Yet, as pointed out by Cinque and Krapova (2007), Bulgarian does allow sentences that call for YEAR(S), e.g.:

(27) Az sâm na petdeset. ('I am of fifty' = 'I am fifty years old')

Their proposal is that Bulgarian patterns here with English, and against French and Italian, because Bulgarian is like English in using *be* with simple expressions of age (as seen in (27) itself), whereas French and Italian use *have*, as in:

(28) Marie a sept ans. ('M has seven years')

(29) Maria ha sette anni. (same)

Cinque and Krapova's idea that a language can use YEAR(S) with expressions of age only if it uses (its counterpart of) *be* in sentences like (27)-(29) is of substantial interest despite having to immediately confront the contrast between (25)/(26) and their English counterpart:

(30) At the age of seven, Mary...

the problem apparently being that there is no visible verb at all in (25)/(26)/(30).

Although the detailed execution of their idea remains to be worked out, it will likely be facilitated by the following proposal, which now attributes to (30) a silent BE(ING):

(31) at the age of BEING seven, Mary...

5. Cartography and English *well*.

The preceding discussion of *inning*/INNING and *year*/YEAR was more relevant to the question of antecedentless silent elements than it was to the question of cartographical hierarchies. Let us now turn to another aspect of English syntax that bears simultaneously on the two questions.

Consider what we think of as adverbial *well*, in sentences like:

(32) You may well be right.

⁵For details on the singular vs. plural question, see Kayne (2003).

From a cartographic perspective we would ask where exactly in the sentential hierarchy this *well* should be placed. Yet I will now suggest that there is no answer, i.e. that *well* may have no place in the sentential hierarchy at all, once antecedentless silent elements are brought into the picture.

Let us begin with the fact that this instance of *well* is incompatible with almost all modifiers, e.g.:

- (33) *You may extremely/exceptionally/fairly well be right.
- (34) *You may so well be right that you've become overconfident.
- (35) *You may not all that well be right.
- (36) *Of course you may this well be right.
- (37) *How well may I be right?

With the further fact in mind that *well* and *good* have in general a lot in common in English, we can now home in on a surprising fact about a very particular instance of *good*, that of:

- (38) You obviously have a good many friends.

namely that this instance of *good* is also incompatible with modification:⁶

- (39) *You obviously have an extremely/an exceptionally/a fairly good many friends.
- (40) *You have such a good many friends that...
- (41) *You clearly don't have that/this good a many friends.
- (42) *How good a many friends do you have right now?

The strong parallelism between (33)-(37) and (39)-(42) gives plausibility to the following proposal. Just like *good* in (38), *well* in (32) is not adverbial, i.e. not in the sentential hierarchy at all, but is rather a DP-internal modifier of (a silent counterpart of) *many*, as shown as a first approximation in:

- (43) you may well MANY be right

Of course *many*/MANY needs a noun (which I now add):

- (44) you may well MANY WAYS be right
-

⁶There is a contrast between:

- i) You may very/(?)perfectly well be right.

and:

- ii) You obviously have a ??very/*perfectly good many friends.

that remains to be understood, as does the contrast between:

- iii) You may as well admit that you're wrong

and:

- iv) *You obviously don't have as good many friends as we do.

The *very* of (i), which may or may not be a functional element, recalls that of:

- v) the very/*extremely same book

- vi) his very/*extremely own car

A reviewer finds the following impossible:

- vii) You may even well be right.

(I find it marginal but not sharply out.) For those who reject (vii), the question of the status of:

- (viii) You may even in a good many ways be right.

comes to the fore.

which in turn almost certainly calls for a preposition (and, as in (38), an indefinite article), yielding finally:

(45) you may IN A well MANY WAYS be right
which is intended to mimic the sentence:

(46) You may in a good many ways be right.

In conclusion, the analysis indicated in (45), with its various antecedentless silent elements, provides an alternative to an analysis of (32) that would have *well* be an adverbial modifier, and in so doing, sharply changes the cartographic picture of such sentences.

We might now ask how general we would expect this analysis of *well* to be. A theoretically restrictive answer, correlated with the anti-homophony position (or principle, or conjecture) put forth in Johns (1992, 84), Embick (2003, 156) and Kayne (2019a), would say that every instance of *well* that is functional (as opposed to lexical, like the *well* that is a source of water) must reflect a single *well*. If so, then we would be encouraged to look in every such case for a silent noun that *well* would modify.

6. Cartography and apparent sentence-final particles.

Consider this English sentence:

(47) We're on the list, right?

One might well think that *right* is, in (47), an instance of a high sentence-final particle (of the sort discussed by Pan (2022) for Chinese) that could be fit into a cartographic sentential hierarchy. Yet, the correct analysis of (47) may rather be as in the following:⁷

(48) we're on the list, ISN'T THAT right?

modeled on the acceptable sentence:

(49) We're on the list, isn't that right?

That is, *right* in (47) may well best be taken to be a reduction, in the sense of (48), of the *isn't that right* seen overtly in (49). If this is on the right track, then *right* in (47) is not in any sense a sentence-final particle; *right* is an adjectival predicate, not to be placed in a high position within a cartographic sentential hierarchy.

A reviewer points out that in addition to (49) English allows:

(50) We're on the list, isn't that so?

yet does not allow in anything like the same sense:

(51) *We're on the list, so?

In other words, the question is whether the analysis given in (48) overgenerates.

Possibly. But even if it does, there would be no advantage to a sentence-final particle analysis of (47), which would have to stipulate that *right* can be a SFP, as opposed to the apparently similar *so*.

This similarity, though, may be exaggerated, thinking of the close relation between (49) and:

(52) Aren't we right to think that we're on the list?

along with the fact that (50) does not map to a corresponding well-formed sentence, as shown by the deviance of:

(53) *Aren't we so to think that we're on the list?

⁷Cf. Kayne (2016).

The contrast between (52) and (53) suggests that we might take (47) to actually be as in:

(54) we're on the list, isn't that right OF US
thinking in turn of:

(55) Wasn't it right/*so of us to say that?
An analysis on the order of (54), like the earlier proposal in (48) that it promises to supersede, will of course preserve the conclusion that *right* in (47) is not a SFP.

7. The hidden complexity of modals.

An antecedentless silent HAVE played a key role in the Harves and Kayne (2012) analysis of English verbal *need*, which, in the spirit of Hale and Keyser (1993; 2002), took verbal *need* to lend itself to an analysis whereby a nominal *need* is paired with a silent light verb HAVE. If we now turn to English modal *need*, as found in sentences like:

(56) You needn't leave so soon.
an initial cartographic account might take this modal *need* to straightforwardly fit into the IP area of a cartographic sentential hierarchy.

Alternatively, following Kayne (2024a), we can, still much in the spirit of Hale and Keyser, take English modal *need* to reflect the combination of nominal *need* with a silent modal (rather than with the HAVE appropriate for verbal *need*), thinking in particular of archaic:

(57) You must needs do that.
a current counterpart of which would be:

(58) You must of necessity do that.

Taking (57) as a clue, this alternative proposal for modal *need* is as in:

(59) ...n't MUST need...
with nominal *need* here being related to silent MUST, not as a direct object, but rather in the way that adjunct-like, non-direct-object *necessity* is related to must in (58), possibly with a silent OF:

(60) ...n't MUST (OF) need...

Movement of this non-direct-object *need* past MUST and *n't*, will yield what we think of as modal *need*,⁸ though contrary to what might have been thought, it is only MUST,⁹ not *need*, that is part of the cartographic sentential hierarchy.

8. PUT and silent agents.

On the basis of auxiliary selection differences between French and Italian, Kayne (2009) attributed both a silent predicate CAUSE and a silent (non-agentive) causer to sentences like:

(61) The ice melted.

A silent CAUSE, though in the following example accompanied by a silent agent phrase, would seem in a similar way to be appropriate for sentences like:

(62) The flowers are in the kitchen on purpose.

with an analysis along the lines of:

(63) the flowers are PUT in the kitchen on purpose

with *on purpose* in (62) now related to silent PUT, and to the understood agent¹⁰ of silent PUT, rather than being related directly to the overt copula. In other words, *on purpose* in (62) is, despite appearances, not part of any sentential hierarchy associated primarily with *are*, given the presence of (antecedentless) PUT.

Both (62) and (61) have in turn something in common with a phenomenon called to my attention in this context by Marcel den Dikken (p.c.),¹¹ namely the existence of verbs that are unaccusative (to judge in particular by auxiliary selection in Italian), yet allow an agentive interpretation for their subject, as in the following:

(64) They arrived late on purpose.

(65) They intervened strongly in the debate.

If the argument that ends up in subject position in such sentences starts out as an object-like argument of the unaccusative verb, how is an agentive interpretation licensed?

⁸On incorporation of adjunct-like elements, cf. English compounds such as *head-fake*, *need-based*, *backflip* et al. The extent to which languages other than English might have recourse to incorporation of (their counterpart of) adjunct-like *need* to (their counterpart of) a silent MUST remains to be determined, one relevant question being whether or not the highly specific properties of English modals might facilitate such incorporation.

Stephanie Harves (p.c.) asks whether the deontic vs. epistemic distinction that holds with *must* carries over to *need*. It may, given the contrast:

i) He needn't work so hard.

ii) There needn't be any solution to that equation.

As Jean-Yves Pollock (p.c.) points out, the expectation is that the epistemic MUST that would presumably be (silently) present in (ii) would be compatible with *necessity*, as does seem to be plausible, given:

iii) There must of necessity not be any solution to that equation.

⁹On OUGHT (and SHOULD), cf. Kayne (2022).

¹⁰On the syntactic representation of which, see Collins (2005, sect. 6; 2024).

¹¹Who points out the relevance of Zaenen (1993).

A solution that comes to mind, in the spirit of the discussion of (62) and (61), is to take (64) and (65) to be significantly similar to:

(66) They made themselves arrive late on purpose.

(67) They made themselves intervene strongly in the debate.

That is, we should associate with (64) and (65) derivations involving both a silent MAKE/CAUSE and an agent of MAKE/CAUSE, in which case *on purpose* in (64) will be linked to that agent phrase (rather than to the argument of *arrive*), in a way that indirectly supports the presence of PUT itself in (62).

9. A complex syntax for malefactive.

In another area of English syntax we find:

(68) Their dog died on them yesterday morning.

with *on them* appearing at first glance to belong to a simple sentence whose main predicate is *died*, and appearing, since *on them* is not a complement of *die*, to be in some high Spec position of the cartographic sentential projection line.

Alternatively, though, we might want to analyze (68) in a way that brings it closer to the following:

(69) Their dog's death hit them yesterday morning.

and even closer to:

(70) ?Their dog's death came down/fell on/upon them yesterday morning.

This would amount to taking (68) to contain a tenseless small clause *their dog die* parallel to *their dog's death* in (70), as in, for example:

(71) -d FALL [their dog die] on them yesterday morning.

in which *on them* is now a locative-like complement of silent FALL.

In (71), the past tense *-d* goes (as it does in (70)) interpretively speaking with FALL, whose silence allows the (mistaken, if I'm right) impression that *-d* in (68)/(71) is interpretively speaking associated directly with *die*.¹² As the derivation of (68) continues, starting from (71), in which *their dog die* is the theme argument of unaccusative FALL, the two subparts of *their dog die* will raise (probably separately) past *-d* (and past FALL),¹³ with *the dog* ending up in any event in matrix subject position.

10. Adjectives modifying silent nouns.

All of the preceding have something in common, I think, with what we can take to be a further case in which the presence of silent elements affects the cartographic picture. Let me begin with the subcase of color adjectives, which might be thought to lend themselves to being placed directly into a DP-internal cartographic hierarchy associated with a pronounced lexical noun, as in:

(72) They just bought a blue car.

in which *blue* would be easy to take to be a modifier of lexical *car*.

However, the proper analysis of this kind of sentence is more likely to be as in:

¹²Whether this sort of analysis is appropriate for any of the various datives found in Romance languages needs to be looked into.

¹³If *die* raises past *-d* as a head, rather than as a remnant, an HMC question arises, on the assumption that the HMC sees silent elements.

(73) They just bought a blue COLOR car.
in which *blue* is a modifier of silent COLOR (as it can be of pronounced *color* in relevant sentences), rather than of *car*. If so, then *blue* is not itself directly in such a DP-internal hierarchy (although the phrase ‘blue COLOR’ could be).

Other adjectives call, I suspect, for comparable analysis. For example, *small* in:

(74) They just bought a small car.
could well be analyzed as in:

(75) they just bought a small SIZE car.
with SIZE in (74)/(75) strongly parallel to COLOR in (72)/(73), and with SIZE here supported by sentences like:

(76) He just bought a size ten pair of shoes.

Probably a bit less close to COLOR than SIZE is, yet still fairly close, is the silent noun NUMBER of sentences like:

(77) They have few friends.
Although *few* in (77) is adjectival, as indicated by the existence of the forms *fewer*, *fewest*,¹⁴ there is good reason to take (77) to be as in:

(78) they have few NUMBER friends.
in which *few*, though still an adjective, is a modifier of NUMBER, not of *friends*. The analysis in (78), which is supported by the relative acceptability of:¹⁵

(79) Of all our students, John’s the one who’s written the fewest number of papers this year.

implies that *few*, contrary to initial appearances, cannot be part of the DP-internal hierarchy associated in (77)/(78) with *friends*.

The presence of silent COLOR, SIZE, NUMBER in the preceding is not to be thought of as an accidental fact of English. Instead, we should be thinking in terms of what I called in Kayne (2005a, Appendix) a Principle of Decompositionality, stated there as:

(80) UG imposes a maximum of one interpretable syntactic feature per lexical item. The idea was and is that no lexical item can be so complex, interpretively speaking, as to simultaneously introduce, as in, say, (72), both the notion of color and the notion of a specific portion of a color scale. The adjective *blue* can pick out a portion of the scale, but color itself must be introduced separately, by the noun *color* or its silent counterpart COLOR.

11. KIND.

Another example of this general sort, i.e. of an adjective not modifying the noun that it at first appears to modify, with implications for what cartographic hierarchy it might be found in, can be seen in:¹⁶

¹⁴As noted by Jespersen (1970/1914, 106). Unlike *few* (and *little*), *many* and *much* are likely not adjectival - cf. Barbiers (2007, 869) and Kayne (2024b).

¹⁵I have even heard a native speaker of English use the phrase *a few number* (which is not possible for me). The generalization seems to be that, with overt *number*, *few* is best if raised, in particular in the manner of superlatives; for discussion, see Kayne (2005a).

¹⁶Cf. Kayne (2019a).

(81) There will be three different wines at dinner later today.
which readily lends itself to the following analysis:

(82) there will be three different KIND wines at dinner today.
in which *different* is more closely associated with silent KIND than with pronounced *wines*. (Future work will need to address the question of how extensive the available set of antecedentless silent nouns like COLOR, SIZE, NUMBER, KIND is, as well as the question of the extent to which languages differ in their use of them, not to mention the familiar question of what exactly their licensing conditions are.)

The postulation of silent KIND has additional motivation, as can be seen, for example, from the following:¹⁷

(83) [the same KIND] eyes
which seems to be called for for an understanding of the usual interpretation of:
(84) John has the same eyes as his mother.

12. Focus-preposing and clefts

In the spirit of all the preceding discussions, one might wonder about the degree of similarity between focus-preposing and cleft sentences, with the possibility existing that focus-preposing necessarily involves a matrix copula (either silent or pronounced) comparable to that seen in clefts, rather than a FocusP in the cartographic CP area in exactly the sense of Rizzi (1997).

That focus preposing might consistently involve a bisentential structure is suggested by what Ofori (2011) says about Akan: “The basic focus marker, we argue, is an equational copula/verb that joins a focused-unit and a presuppositional-constituent”.¹⁸

Anna Szabolcsi (p.c.) has made a similar suggestion (based on work by Soo-Hwan Lee¹⁹) regarding Korean NOM(inative)-marked items, noting that if focussed NOM-marked phrases were actually parts of clefts, then the seemingly dual role of NOM in Korean (as Case-marker and as focus-marker) would dissolve and become compatible with anti-homophony.²⁰

Guglielmo Cinque (p.c.) has noted that Italian focus-preposing and Italian clefts differ with respect to negative phrases, which can be subject to focus-preposing, but cannot be clefted. This would appear to be unexpected, if focus-preposing involves a cleft-like derivation. However, Pollock (2021) notes that French interrogatives having the form of interrogative clefts, as in:

(85) Qu'est-ce que tu fais? ('what is it that you are-doing')

¹⁷As well as from:

i) ...blue one KIND CAR -s
as an analysis of *blue ones*:
ii) I prefer red cars, but you prefer blue ones.
For relevant discussion, see Kayne (2017).

¹⁸Cf. also Titov (2019).

¹⁹Cf. Lee (2024).

²⁰If focus-preposing is invariably associated with a cleft-like structure involving a copula, one will need to ask why the language faculty would have taken that path. Relevant may be den Dikken (2021), in particular if his 'relator' can be identified with the copula in question; cf. also den Dikken (2006).

“despite their bi-clausal forms would best be rendered as simple mono-clausal Wh-questions in English...none have a cleft interpretation”. Thus it might be that what we call focus-preposing is invariably, in whatever language, akin to French (85) in that respect.

13. An analysis of right-dislocation as bisentential.

Indirectly related to the question of focus-preposing is the proposal I made in Kayne (1994, sect. 7.3), to the effect that right-dislocation of the sort seen in:

(86) He's real smart, John.

could be akin to sentences like:²¹

(87) He's real smart, John is.

That is, (86) might contain a silent copula parallel to the pronounced *is* of (87). If so, then *John* in (86) would not itself cartographically speaking be in the Spec of any projection in the CP area of *He's real smart*, i.e. would not be associated with any part of the sentential hierarchy associated with *he's real smart*.²²

The question then arises as to what the proper analysis of (87) (and arguably by extension (86)) is. Since some English seems to allow *so* to appear between the two halves of (87), let me make as a side proposal:

(88) he's real smart SO John is

It certainly looks like deletion (in this case probably VP-deletion) has applied in the second half of (87)/(88). But if that were the whole story, how would we be able to block the following:

(89) *He's real smart, John is in Paris.

in which the identity condition for deletion is not met? And similarly for this contrast:²³

(90) He likes phonology a lot, John does.

(91) *He likes phonology a lot, John often goes to Paris.

14. High adverbs

Finally, let me touch on the question of high adverbs such as *unfortunately*. The possibility exists of relating the following two sentences:

(92) Unfortunately, you're late.

²¹For monosentential analyses of right-dislocation compatible with antisymmetry, see Cecchetto (1999) and Villalba (1999).

²²Whether this approach to right-dislocation can be extended to Cinque's (1990) CLLD remains to be seen; cf. Ott (2014).

²³A possible answer is that movement is involved, starting from a derivational stage that looks like:

(i) J does he likes phonology a lot.

with subsequent obligatory high movement of the phrase 'he likes phonology a lot' past both *John* and *SO* and with further questions arising that are beyond the scope of this paper, as is the question of how best to extend a movement-based bisentential approach to:

ii) They said they would do it and do it they have.

a type of sentence discussed recently in partially different terms by Ott (2018, sect. 3.4) and Thoms and Walkden (2018, sect. 2.3).

(93) It's unfortunate that you're late.
which differ, however, with respect to wh-movement:

(94) *How unfortunately are you late?

(95) How unfortunate is it that you're late?

which reduces the plausibility of deriving (92) from anything like (93).²⁴

An alternative approach to (92) would attempt to link it to:

(96) The unfortunate thing is, (is) you're late.²⁵

in which case, in a way partially similar to Ott (2014), *unfortunately* would not be in a high cartographic Spec position of the sentence *you're late* in (92) any more than *the unfortunate thing* is in (96). The impossibility of (94) might then (if *it* is set aside) be related to that of:

(97) *How unfortunate the/a thing is, (is) he's late?

Other high adverbs may also best be analyzed as in (96). For example, *hopefully* shows the same restriction as in (94), i.e. we have:

(98) Hopefully, everything will work out.

but not:

(99) *How hopefully will everything work out?

If we set aside the (challenging) question of *-ly*, as with *unfortunately*, and also that of the suffix *-ful-*, (98) could be understood as close to:²⁶

(100) My hope is that everything will work out.

or perhaps to:

(101) The thing to be hoped is that everything will work out.

with (99) then akin to:

(102) *What hope of yours is that everything will work out?

or to an impossible wh-variant of (101).

15. Conclusion

In conclusion, our understanding of cartographic hierarchies must go hand-in-hand with our understanding of the distribution and properties, in a given language and cross-linguistically, of silent elements of the various sorts discussed above.

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²⁴For relevant discussion, see Nilsen (2003).

²⁵Massam (1999, 338) suggests, for sentences of this sort, the possibility of a silent operator:

(i) WHAT the thing is, is...

²⁶Similar challenges arise for *importantly*, *arguably*, *thankfully*.... One will also want to account for the absence, as high adverbs, of **rectangularly*, **earlyly*, **farly*....

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